Nicholas M. Galus

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GitHub Pages

Objective

Motivated Computer Engineering student with experience in embedded systems, digital hardware design, and control logic development. Skilled in microcontroller programming, FPGA design, and system-level debugging using C, Verilog, and engineering test tools. Seeking opportunities to contribute to real-time hardware–software solutions across innovative technology fields including automotive, defense, and industrial automation.

Technical & Engineering Skills

Programming & Embedded Systems

- C, C++, Python, Java, x86 and ARM Assembly, Verilog, MATLAB, HTML/CSS
- Raspberry Pi and Arduino for embedded prototyping and control systems, including OpenPLC and mySCADA development environments
- Git, GitHub, Linux, VS Code, VIM, Eclipse, GCC and GDB for compiling and debugging C applications

Hardware Design, Testing & Analysis

- Digital logic design and FPGA development using Verilog RTL and Xilinx tools for synthesis and timing simulation
- Oscilloscope, logic analyzer, and multimeter operation for signal verification and circuit debugging
- Hardware prototype validation with soldering, breadboarding, and digital logic testing instrumentation

Certifications

- Java – Information Technology Specialist
- SolidWorks CAD Design Associate
- Microsoft Office Specialist (Excel, Word, PowerPoint)

Education

Bachelor of Science in Computer Engineering — **Anticipated: May 2028**

St. Mary's University, San Antonio, TX

Professional Experiences

Research & Development Intern - San Antonio Museum of Science and Technology (SAMSAT) — **Fall 2025 to Present**

- Refined MyBox system to improve communication between the web interface and Raspberry Pi controllers managing lighting
- Created technical documentation enabling on-site technicians to diagnose and repair hardware and software issues independently
- Optimized infrastructure layout to reduce cable clutter and added animated visuals simulating city systems and data flows

Cyber Engineer Intern - ICR, San Antonio, TX — **Summer 2025**

- Developed and optimized C and x86 Assembly applications to support software testing and performance validation
- Collaborated with engineers to test, debug, and document code for system-level reliability and accuracy
- Presented project outcomes at the end of the summer, highlighting key contributions and technical skills gained

Undergraduate Researcher - Spring Research Fellowship — **Spring 2025**

- Modeled permafrost thaw dynamics using coupled Budyko energy balance and heat transfer equations to study climate feedback
- Simulated system behavior using finite-difference and Euler methods in Python to compare proposed coupling mechanisms
- Presented research at the 2025 Texas Undergraduate Mathematics Conference (TUMC) and the St. Mary's Spring Research Showcase

Intern - City of San Antonio (IT Services) — **Summer 2024**

- Processed and documented 600+ devices to improve recycling, tracking, and lifecycle management systems
- Provided technical support for Android and iOS devices, assisting 10-15 users daily with troubleshooting and setup

Apprentice in Training - Accenture Federal Services (IT Technical Support) — **Spring 2024**

- Deployed and configured 30+ Windows laptops with verified software installation and performance testing
- Resolved 75+ Technical Support Tickets and assisted senior technicians with hardware and network diagnostics

Intern - City of San Antonio (Public Works) — **Summer 2023**

- Analyzed traffic data from 2,000+ traffic signals and generated 50+ technical reports to identify optimization needs
- Programmed and synchronized 30+ traffic signal controllers to enhance citywide traffic flow and reduce congestion

Leadership & Involvement

- **Society of Hispanic Professional Engineers (SHPE)** – Member (2025 – Present)
- **IEEE Student Branch** – Treasurer (2025 – Present); Member (2024 – Present)
- **President's Ambassador** – University Representative (2025 - Present)
- **Honors Program** – Leadership Council Mentor (2025 - Present); Honors Scholar (2024 - Present)
- **Resident Assistant** – Student Leader (2025 - Present)

Selected Projects

- **Protein Dispenser** – Designed and 3D-printed a dispenser that improves protein serving efficiency and keeps usage clean and simple
- **VibePulse** – Developed a mobile app that monitors heart rate and automatically adjusts music tempo and song selection based on activity